DSM2274N/2294N System Administration & Management

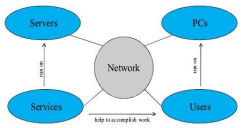
Lec01 - Introduction

Topics

1. What is System Administration?
2. What do system admins do?
3. Principles and First Steps
4. Organizations and Certifications
5. Maturity and Complexity
6. Ethics

What is a System?

- System: An organized collection of computers interacting with a group of users.



What is System Administration

- System administration is the field of work which involves...
  - Managing one or more systems, in their various states
  - System could be software, hardware, Servers or workstations.
  - The objective are efficiency and effectiveness
- System administration matters because computers and networks matter.

  Business Operations | Colloborations | Solve Many Problems
- Business Depends on

  Internet | Intranet | World Wide Web
- Management now has a more realistic view of computers, More people becoming direct users of Computers
- Unreliable machine room power system that caused occasional but bearable problems now prevents sales from being recorded.
- Computers matter more than ever. If computers are to work and work well, system administration matters

Orgnization Size and System Administrtaion

- Small
  - **2-10** computer's 💻, **1** OS, **2-20** user's 👥
  - The system administrator (sys-admin) can be entire IT staff
  - Small staff size may require outsourcing to obtain most specialized skills.
- Mid-sized
  - **11-100** computer's 💻, **1-3** OSes, **21-100** user's 👥
  - These organizations are larger than small businesses but smaller than large enterprises.
- Large
  - **> 100+** computer's 💻, **multiple** OSes, > **100+** user's 👥
  - The sys-admin is one of many IT staff, a specialist instead of "jack of all trades"
  - Other IT roles may include database admin, network admin, fileserver admin, help desk worker, programmers, logistics
  - Outsourcing is done to reduce cost, and for some specializations.

What is a System Administrator?

1. Difficult to define because system administrators do so many things:
   - looks after computers, networks, and the people who use them.
   - look after hardware, operating systems, software, configurations, applications, or security
2. A system administrator influences how effectively other people can or do use their computers and networks.

   Help Employees | Support Others | Guide Others
3. A system administrator sometimes needs to be a

   business-process consultant | corporate visionary | janitor | software engineer | electrical engineer | psychiatrist | mindreader | economist
4. A role that is known by different names

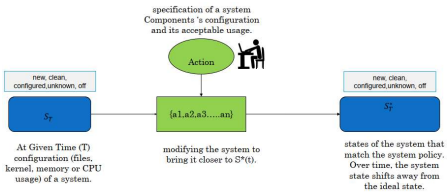
   network administrators | system architects | system engineers | operators | system programmers

What does Managing an IT System means?

- Applying Tasks and actions:
  - Installation and Configuration
  - Monitoring, Logging and Reporting
  - Changing
  - Supporting
  - Provisioning
  - Troubleshooting

Changes System State

Formal Definition of System Administration



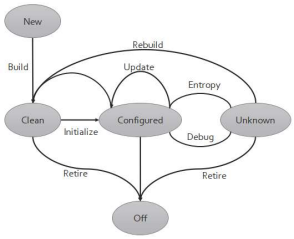
System State

- **System Policy**: specifications of a system's configuration and its acceptable usage.
- **System State S(t)**: the current configuration (files, kernel, memory or CPU usage) of a system.
- **Ideal State S*(t)**: States of the system that match the system policy. Over time, the system state shifts away from ideal state.
- **System Administration**: Modifying the system to bring it closer to S*(t).

- **Evard's Computer Machine Life Cycle:**
  - A Computer Machine Goes through different states in their life.
  - Different Processes are used for transition of the state of Computer Machines
  - Evard has outline different states and process that are performance on

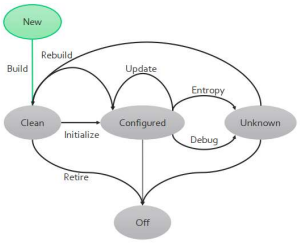
Evards ' Life Cycle Of Computer Machine

- Five States
  - New
  - Clean
  - Configured
  - Unknown
  - Off
- • Several Processes
  - Build
  - Initialize
  - Update
  - Entropy
  - Debug
  - Retire



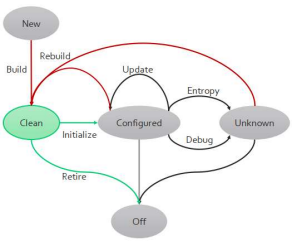
Computer Machine Life Cycle – New State

- **New State:**
  - A new machine
  - Unpacking
  - Physical Connections
  - Power up
  - Preinstalled OS or No OS



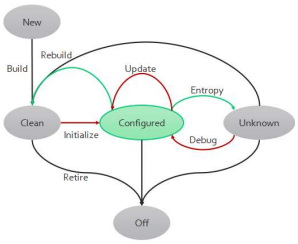
Computer Machine Life Cycle – Clean State

- **Clean State**
  - A machine that has the operating system installed but isn't set up to function in the System.
  - **Build Process**
    - Select OS
    - Install OS
    - Drivers and Software
  - **Rebuild Process**
    - Recovered Data
    - Reformat Disks
    - Reinstall OS/ Software



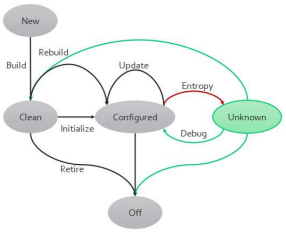
Computer Machine Life Cycle – Configured State

- **Configured State**
  - A machine that is properly configured to meet the needs of the System.
  - User Roles and Permissions
  - Updates
  - **Update Process**
    - Select Updates
    - Schedule Updates
    - Monitor



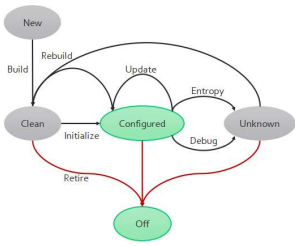
Computer Machine Life Cycle – Unknown State

- **Unknown State**
  - A machine that has been misconfigured, or that has become out of date, or that has been borrowed by an intern and returned stained
  - **Debug Process**
    - Troubleshooting
    - Remove the bugs
    - Fix Drivers and Software
  - **Entropy Process**
    - Changes by user/admin/updates
    - Compromised State
  - **Rebuild Process**
    - Recover User Data
    - Format Disks and Reinstall OS



Computer Machine Life Cycle – Off State

- **Off State**
  - Powered Off
  - No User Data
  - OS installed or Removed
  - Disconnected from Infrasture
  - **Retire Process**
    - Recover User Data
    - Flush the Hard Disks
    - Isolate from Infrastructure



System Administration Maturity Model (SAMM)

- Five SAMM levels:
  1. Ad Hoc
     - Ad-Hoc non-repeatable solutions, firefighting.
  2. Repeatable
     - Some repeatable process
  3. Defined
     - Documented standard process
  4. Managed
     - Process effectiveness measured, adapted.
  5. Optimized
     - Continuous Improvement

Level 1 Ad-Hoc – Common Practices

- **New user** – Verbal requests are addressed as time permits or escalated with management.
- **Software install** – New or upgraded software is installed whenever & where ever makes the most sense at the moment.
- **Hardware install** – New or upgraded hardware is installed whenever & where ever makes the most sense at the moment.
- **Problem report** – Problems are sometimes reported by users by mail or phone to a random administrator.
- **Security** – No specific security standards or policies exist.
- **Disk capacity** – Disk space is in short supply. No information on usage rates is available. Project managers of supported organizations often fight among themselves regarding disk space.
- **Backups** – Backups are usually done according to a weekly schedule.

Level 2 Repeatable – Common Practices

- **New user** – Procedure to request and create an account is well documented. Cycle time for requests is monitored.
- **Software install** – Installation guidelines are understood. Time spent installing and configuring software is tracked.
- **Hardware install** – Installation standards are understood. Time spent installing and configuring hardware is tracked.
- **Problem report** – Process to report problems is well understood by users and cycle time for problem resolution is monitored.
- **Security** – Various security standards are clearly documented. Security violations are monitored.
- **Disk capacity** – Acceptable disk capacity levels are established. Capacity is periodically monitored.
- **Backups** – Failure conditions for backups are understood. Failure rates and effort to resolve problems are tracked.

Level 3 Defined – Common Practices

- **New user** – Head count expansion information is provided to signal new account requests are expected. Revision of procedure.
- **Software install** – Installations are planned with the supported organizations. Reasons for install/upgrade are recorded.
- **Hardware install** – Installations are coordinated with supported organizations and vendors. Reasons for install/upgrade are recorded.
- **Problem report** – Problems are mapped to root causes. Group reviews solutions to resolutions suggested to address root causes.
- **Security** – Group reviews security incidents for root vulnerabilities. Resolutions are discussed, tested and released.
- **Disk capacity** – Capacity planning is addressed in project plans written by supported organizations and reviewed by the network & systems group.
- **Backups** – Training is provided for network and systems group to enhance backup programs and procedures.

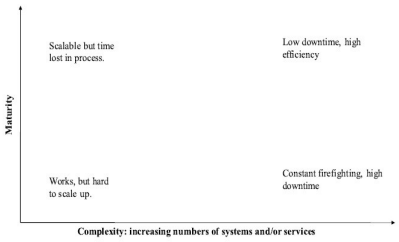
Level 4 Managed – Common Practices

- **New user** – Cycle time numbers are used to adjust staffing to meet demand and requirements.
- **Software install** – Productivity measures and goals for installation created.
- **Hardware install** – Productivity/problem data is compare to goals for installs, tests and demos.
- **Problem report** – Cycle time and resolution quality information is used on a regular basis to access effectiveness of problem reporting and resolution system.
- **Security** – Effectiveness of group reviews is studied.
- **Disk capacity** – Metric for disk availability is created and used.
- **Backups** – Backup system is certified with high reliability rating.

Level 5 Optimized – Common Practices

- **New user** – Accounts are electronically requested and verified.
- **Software install** – Problems with software installs are documented and avoidable with new procedures.
- **Hardware install** – New hardware technologies are evaluated & integrated.
- **Problem report** – New problem reporting system installed to better meet user requirements for ease of use.
- **Security** – Internal contest to establish better security practices is established.
- **Disk capacity** – Project tracking information combined with metrics of utilization are used to predict needs.
- **Backups** – Backup process is revisited with input from supported organizations re: production schedules.

Maturity and Complexity



Maturity Model Level Tools

- Ad-Hoc : OS GUI, CLI, or web administration interfaces
- Repeatable : Version Control (RCS, SVN, GIT), request tracker.
- Defined : Automatic monitoring (Nagios, Monit, god)
- Managed : Configuration management (AutomateIT, cfengine)
- Optimized: Evidence & observation recording, security checks integration (SAST/DAST)

Common Sys-Admin Tasks

1. Manage users (background stuff)
2. Manage hardware
3. Perform backups
4. Install new software systems
5. Troubleshooting
6. System monitoring
7. Auditing security
8. Help users
9. Communicate

User Management

- Creating user accounts
  - Consistency requires automation
  - Startup (dot) files
- Namespace Management
  - Usernames and UIDs
  - Multiple namespaces or SSI (Single Sign In)?
- Removing user accounts
  - Consistency require automation
  - Many accounts across different systems

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Hardware Management

- Adding and removing hardware
  - Configuration, cabling, etc.
- Purchase
  - Evaluate and purchase servers + other hardware
- Capacity Planning
  - How many servers? How much bandwidth, storage?
- Data Center Management
  - Power, racks, environment (cooling, fire alarm)
- Virtualization
  - When can virtual servers be used vs. physical?

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Backups

- Backup Strategy and policies
  - Scheduling: When and how often?
  - Capacity Planning
  - Location: on site vs offsite
- Monitoring backups
  - Checking logs
  - Verifying media
- Performing restores when requested.

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Software Installation

- Automated consistent OS installs
  - Desktop vs Server OS image needs.
- Installation of Software
  - Purchase , find, or build custom software.
- Managing software installations
  - Distributing software to multiple hosts.
  - Managing multiple version of software package.
- Patching and updating software.

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Troubleshooting

- Problem Identification
  - By user notification
  - By log files or monitoring programs
- Tracking and Visibility
  - Ensures users know you're working on problem
  - Provide an ETA if possible.
- Finding the root cause of problems
  - Provide temporary solution if necessary
  - Solve the root problem to permanently eliminate

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System Monitoring

- Automatically monitor system for
  - Problems (disk full, error logs, security)
  - Performance (CPU, mem, disk, network)
- Provides data for capacity planning
  - Determine need for resources
  - Establish case to bring to management

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Helping Users

- Request tracking system
  - Ensures that you don't forget problems.
  - Ensures users know you're working on their problem; reduce interruptions, status queries.
  - Lets management know what you've done.
- User documentation and training
  - Policies and procedures
- Schedule and communicate downtimes

Communicate

- Customer
  - Keep customer appraised of process.
    - When you've started working on a request with ETA.
    - When you make progress, need feedback.
    - When you're finished.
  - Communicate system status
    - Uptime, scheduled downtimes, failures.
  - Meet regularly with customer managers
- Managers
  - Meet regularly with your manager.
  - Write weekly status reports

Specialized Skills

- Heterogeneous Environments
  - Integrating multiple-OSes, hardware types, or network protocols, distributed sites.
- Database
  - SQL RDMS
- Networking
  - Complex routing , high speed networks, voice.
- Security
  - Firewalls, authentication, NIDS, cryptography
- Storage
  - NAS, SANS, cloud storage
- Virtualization and cloud computing
  - VMware, cloud architectures.

Qualities Of A Successful System-Administrator (Sys-Admin)

- Customer oriented
  - Ability to deal with interrupts, time pressure
  - Communication skills
  - Service provider, not system police
- Technical Knowledge
  - Hardware, network, and software knowledge
  - Debugging and troubleshooting skills
- Time Management
  - Automate everything possible
  - Ability to prioritize task: urgency and importance.

1st Step to Better System-Administrator

- Use request system
  - Customer know what you're doing.
  - You know what you're doing.
- Manage quick request right
  - Handle emergencies quickly
  - Use request system to avoid interruptions.
- Policies
  - How do people get help?
  - What is the scope of responsibility for SA teams?
  - What is our definition of emergency?
- Start every host in a known state.

Principles of A Good System-Administrator

- Simplicity
  - Choose the simplest solution that solves the entire problem.
  - Work towards a predictable system.
- Clarity
  - Choose a straightforward solution that's easy to change , maintain, debug, and explain to other SAs.
- Generality
  - Choose reusable solutions that scale up; use open protocols.
- Automation
  - Use software to replace human effort.
- Communication
  - Be sure that you're solving the right problems and that people know what you're doing.
- Basic First
  - Solve basic infrastructure problems before advances ones.

Professional System Administrator Organizations

- USENIX: Advance Computing Association
- LISA: Large Installation System Administration
- SAGE: System Administration Guild
- LOPSA: League of Professional System Administrator.

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System Administrator Certifications

- CCNA, CCNP, CCIE (Cisco)
- cSAGE (SAGE)
- MCSA (Microsoft)
- RHCE (Red hat)
- SCSA (Sun)
- VCP (VMware)

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Job Description Example - SAGE

- **Novice :** OS familiarity, help desk skills
- **Junior:** Can use system administration tools
- **Intermediate:** Understanding of distributed computing, common servers, automate small tasks, independent action
- **Senior:** Understanding of scaling issues, including capacity planning, solve problems by addressing the root cause, higher level programming abilities, write proposals for purchasing, data center planning, etc.

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SAGE Code of Ethics

- Professionalism
- Personal Integrity
- Privacy
- Laws and Policies
- Communication
- System Integrity
- Education
- Social Responsibility

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SYS-ADMIN Case STUDY: Terry Child

- Network Administrator for SAN Fransisco CCIE
  - who built city's FiberWAN network.
- Terry was only person with router passwords
  - IT department acknowledges knowing that
  - He was on-call 24x7x365 to resolve issues
- Terry refused to give password to boss
  - Cited fears that they would be misused by management, outside contractors.
- What was the right thing for Terry to do?

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Other operational issues

1. An employee is working from home and has trouble joining a video call. How would you help them?
2. Some employees are complaining about their Internet speed. What steps would you take to identify the problem?
3. We would like to create simple manuals to help employees to use our equipment properly. What instructions would you give to help your colleagues use a printer?
4. How would you deal with a virus?

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Key Points

- Defination
  - System, system state, ideal state, administration
- Principles of System Administration
  - Simplicity
  - Clarity
  - Generality
  - Automation
  - Communication
  - Basic First
- System Administration Maturity Model
  - Maturity and Complexity, tools

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